

# Enhancing Tactile Graphics Exploration with Large Language Models

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Tactile graphics (TGs) are essential for blind or visually impaired individuals to perceive the layout of spatially distributed data, yet they cannot convey all attributes, such as small details or surface appearance. Augmenting tactile graphics, providing audio feedback associated to the user's touch can convey additional information, but this often force users to passively receive predetermined descriptions, making targeted information retrieval inefficient and limiting autonomous exploration. Large Language Models (LLMs) can solve these problems, but current integrations remain limited to specific domains, like maps or charts. In this demonstration, we address these constraints by extending the CamIO system with an LLM to enable general conversational exploration of TGs across arbitrary content types. The system combines point-based interaction, where users touch hotspots to receive audio descriptions, with talk interaction to converse with the LLM, allowing users to ask questions about the TG. This approach promotes more adaptive, user-driven exploration and supports richer access to spatial and semantic information in TGs.

CCS Concepts: • **Human-centered computing** → *Accessibility systems and tools*; • **Applied computing** → *Arts and humanities*.

Additional Key Words and Phrases: Art accessibility; Audio-tactile interfaces; Visual impairment.

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## 1 Introduction

Tactile graphics (TGs) are commonly used by people who are Blind or have Low Vision (BLV) to access different types of 2D information, including drawings and charts [16]. Despite their effectiveness in conveying geometric information, TGs cannot always convey all relevant attributes of what they represent. In particular, there is a trade-off between the number of tactile details that can be represented and the TG's graphic intelligibility. Also, other properties, such as color and surface appearance, are inherently difficult to encode in tactile form. Braille can supplement tactile elements with textual information, but space limitations typically restrict labels to short abbreviations, and many BLV people can not read braille [15].

Research has addressed these challenges through augmented tactile graphics that combine physical exploration with audio output [3]. These approaches track user interactions with TG elements and deliver spoken descriptions for each touched location. Various technologies have been explored, including standard [2, 5, 6, 12, 17] and depth cameras [11, 14]. CamIO [1], which we previously developed, follows this approach by identifying touch interactions (*i.e.*, when the user touches the TG) through computer vision and providing corresponding audio feedback.

Despite their advantages, current approaches for augmenting tactile graphics exhibit notable constraints. Users must passively receive predetermined descriptions in sequence, making targeted information retrieval inefficient and cognitively demanding when exploring content-rich TGs. Additionally, these systems offer no mechanism for users to seek elaboration on the provided information or obtain alternative explanations, thereby limiting autonomous

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exploration. Recent research has attempted to address these limitations by integrating Large Language Models (LLMs) into augmented tactile graphics. However, existing approaches have focused on specific application domains, such as tactile maps [9, 13] and tactile charts [7, 8].

In this demonstration, we present an extension of the CamIO system [1] that integrates an LLM to support conversational exploration of TGs in general, without being restricted to specific content types or application domains. By supporting natural language queries, the system allows users to actively request information about the TG and obtain clarifications tailored to their needs. This conversational capability enhances the existing point-based interaction, aiming to tailor the exploration of the TG to each user’s needs.

## 2 The CamIO System

CamIO is a web application that allows users to receive feedback while exploring TGs [1]. To use a TG with CamIO, the TG must first be virtually annotated to identify interactable elements (called *hotspots*), providing descriptions for these elements and an overall description of the TG itself. This annotation process is supported by the CamIO-Creator web application [1]. For example, consider a teacher of people with visual impairments (TVI) preparing tactile materials to support a lesson on the biological cell. As shown in Fig. 1, the teacher can define a new hotspot for the Golgi apparatus, draw its position on a representation of a biological cell, and add a description. The output of the annotation process is a CamIO model, which can be loaded into the system to explore the TG.

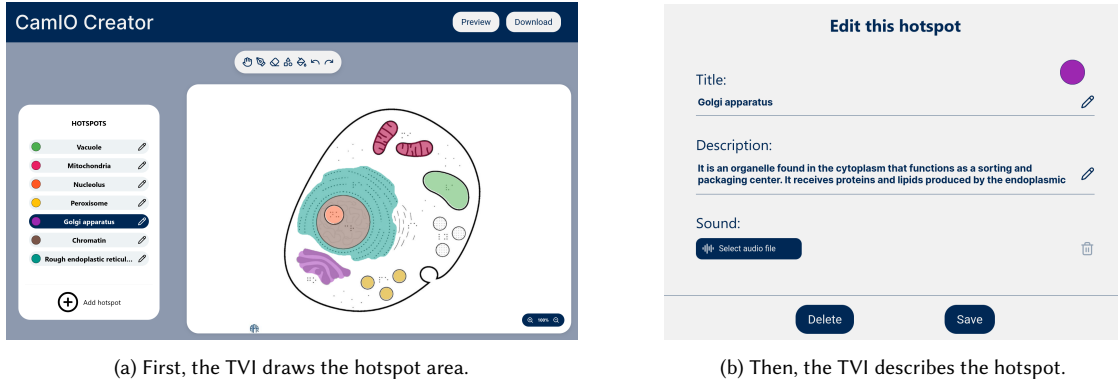


Fig. 1. TVI creates augmented tactile material.

Once a CamIO model is created, a BLV user can use it to access the TG. This requires framing the TG with the device camera. CamIO continuously analyzes the captured video stream to detect which TG hotspots the user is interacting with. This process comprises two stages. First, CamIO localizes the TG within the scene using template-matching techniques. Second, it identifies the user’s hand position and gesture. If the user is pointing at a hotspot, the system then triggers the appropriate feedback. We refer to this interaction as *point inspection interaction*. For example, as shown in Fig. 2a, when the user points to the cell’s Golgi apparatus, CamIO reads the description previously defined by the TVI. Feedback stops when the user moves their finger away from the hotspot. This interaction technique is consistent with prior work [3–6, 9–12, 14, 17] and, in addition to *point inspection interaction*, also allows user to explore the TG without triggering audio feedback, for example, by using the open hand, as shown in Fig. 2b.

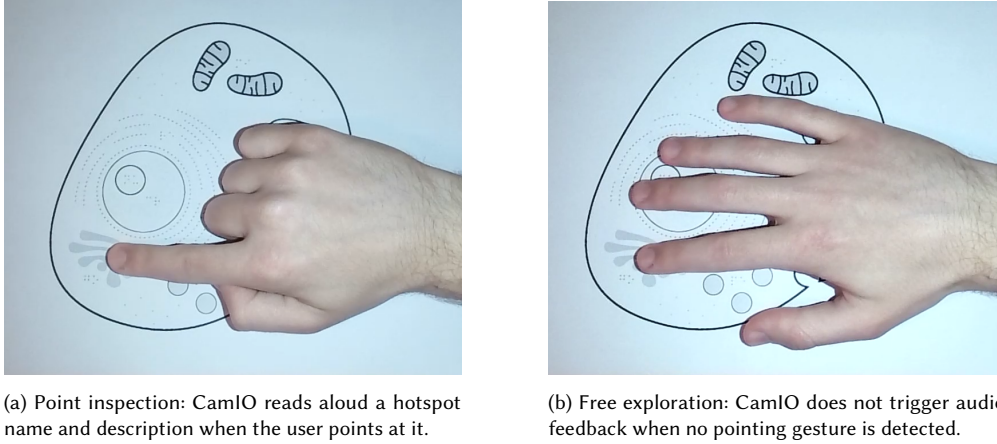


Fig. 2. BVI user exploring a biological cell TG.

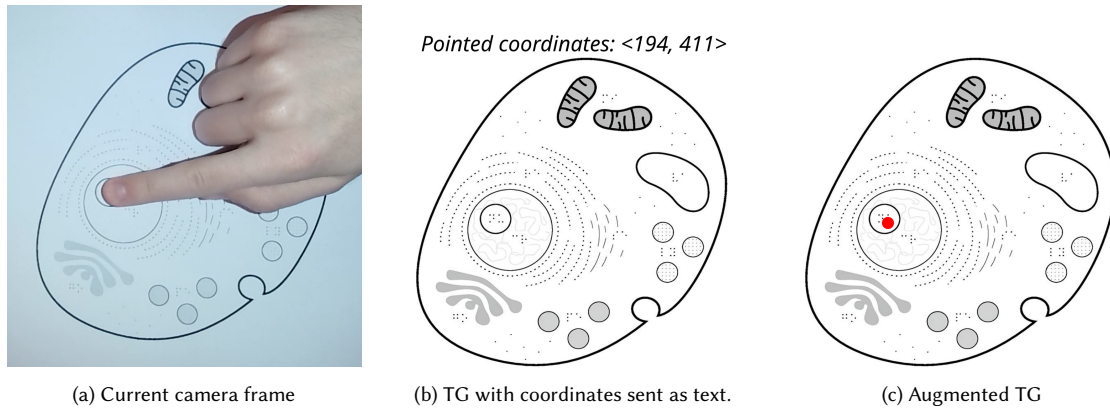


Fig. 3. Strategies experimented with to provide the user's touched position to the LLM.

### 3 Extending CamIO With an LLM

We extended CamIO with a speech-to-speech LLM<sup>1</sup>, allowing users to pose questions about the TG content. We refer to this interaction as *talk interaction*. The extension with LLM required addressing two main challenges: tuning the prompt (Section 3.1) and integrating *talk interaction* into the existing system (Section 3.2).

#### 3.1 Prompt Engineering

Defining the LLM prompt required addressing two specific challenges. The first challenge is to enable the LLM to answer questions based on the current point the user is touching. For example, in the cell drawing, the user may ask “Is this the nucleolus?”. The challenge, therefore, is to convey this visuo-spatial information to the LLM. To enable this, we experimented with different solutions, including providing the LLM with the current camera frame (Fig. 3a), the image

of the TG with the coordinates of the touched point on the TG (Fig. 3b), and the TG augmented with a colored dot where the user is pointing (Fig. 3c).

The second challenge is that the LLM prompt should vary depending on the TG being explored and the CamIO model. Indeed, each model contains the hotspot descriptions that serve as the basis for the information to be provided to the user. For example, when using the cell model, if the user requests information about the nucleolus, the LLM should respond by elaborating on the description associated with the nucleolus hotspot.

To address both challenges, instead of defining a static prompt, we adopt an iterative procedure that processes the CamIO model to generate a model-specific prompt. First, we identified two TGs and their corresponding models, along with a set of 48 benchmark questions for each, partly inspired by previous work [9]. Then, we began with an initial prompt-generating procedure, which we used to generate answers to the benchmark questions. We then manually evaluated the answers, classifying each one into five classes: *wrong*, *partially wrong*, *partial or incomplete*, *correct but not optimal*, and *correct*. By analyzing the errors, we varied the prompt-generating procedure (and possibly the approach to convey geometric information) and repeated the evaluation cycle. We repeated this cycle for a total of 12 iterations. The final prompt-generation procedure results in a prompt that includes a general description of the TG, descriptions of all annotated hotspots, and an image of the TG. The point touched by the user is provided as an image of the TG augmented with a colored dot (Fig. 3c).

### 3.2 Designing Talk Interaction

Integrating *talk interaction* within the existing CamIO system required addressing two main design challenges: how to allow the users to initiate and stop conversations with the LLM, and how to prevent interference between the LLM's spoken responses and the system's existing audio feedback for touched hotspots.

For the first challenge, similarly to existing voice assistants, we defined a wake phrase that activates *talk interaction*. Users say "CamIO start" to begin conversing with the LLM. The system then listens to the user's query and produces a spoken answer. If the user is pointing at the TG while asking a question, the pointed position is included in the query and considered by the LLM in formulating the response. A notification sound alerts the user before the system's answer is played. Users can interrupt lengthy responses by saying "Interrupt response", or they can automatically interrupt the current response by starting a new question. To stop talk interaction, users can say "CamIO stop".

To prevent interference between *talk interaction* and *point inspection interaction*, we designed the following solution: when the LLM begins delivering a spoken response, any ongoing hotspot audio feedback is automatically stopped to avoid overlap. If the user points to a new hotspot during the LLM's response, the system interrupts the LLM's answer and immediately plays the feedback for the newly touched hotspot. This design choice allows users to continue exploring the TG's hotspots while conversing with the LLM.

## 4 Demonstration

This demonstration will allow participants to experience how CamIO integrates an LLM to support conversational interaction during the exploration of a TG.

Three TGs will be available during the demonstration: a biological cell (Fig. 4a), a map of Italy (Fig. 4b), and a representation of the Arc de Triomphe (Fig. 4c). The TGs were selected from the APH Tactile Graphics Image Library<sup>2</sup> based on the complexity of their content. This makes them ideal candidates to showcase the benefits of the LLM

<sup>1</sup>In our prototype, we used gpt-realtime, <https://platform.openai.com/docs/models/gpt-realtime>.

<sup>2</sup><https://imagelibrary.aph.org>

assistance during exploration. For example, when exploring the biological cell, users may ask “What’s the main function of the Golgi apparatus?” instead of listening to the complete hotspot description. For the map of Italy, users can request spatial information, such as “Where is Vatican City?” or “How can I reach Vatican City from my position?”. For the Arc de Triomphe, instead, a question like “What do the two sculptures on either side of the arch represent?” would allow users to understand the symbolic meaning without exploring each sculptural detail individually.

During the demonstration, CamIO will be executed on a laptop connected to an external camera. The camera will be mounted on a tripod to maintain a stable view of the TG. Participants will be provided with headphones to listen to the audio output, in case the environment is noisy.

Each session will begin with a brief introduction to the CamIO system and its two interaction modes: *point inspection* and *talk interaction*. Participants will then freely explore their chosen TGs using both interaction modes. Throughout the demonstration, the supervisor will provide suggestions for effectively interacting with the LLM, such as requesting clarifications on specific elements, requesting elaborations, or seeking information about spatial relationships. This guided support will help participants understand the potential of conversational exploration.

A complete demonstration session will last approximately 5 to 10 minutes per participant.

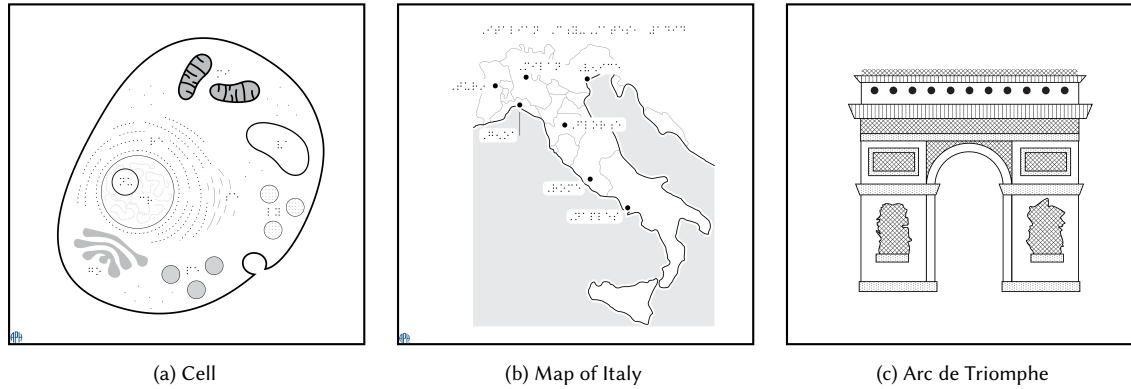


Fig. 4. TGs proposed for exploration during the demonstration.

## 5 Conclusions and Future Work

This paper describes how we extended the CamIO system [1] with a speech-to-speech LLM to enable BLV users to converse with it while exploring TGs. The LLM integrates seamlessly with the previous version of CamIO, allowing users to freely switch between *point inspection interaction* and *talk interaction*. The system enables users to ask questions about the TG’s content, facilitating access to targeted information. In addition, the LLM can provide alternative explanations and expand on hotspot descriptions, allowing users to customize their exploration experience.

The system’s main limitations stem from the LLM itself. LLMs still suffer from hallucinations and may respond incorrectly to user queries. This may lead to misinformation, as users may have no way to verify the LLM’s responses when exploring unfamiliar TGs.

As future work, we plan to conduct a user study with BLV participants to evaluate the user experience and the system’s effectiveness compared to traditional *point inspection interaction* alone, and to identify additional limitations that may emerge during real-world usage.

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